

1 Defining the LUTs

For reducing the computational burden, functions such as $\sqrt{x}$ and $\text{tg}^{-1}(x)$ were implemented through the use of look-up tables (LUTs). Nonetheless, tabling these functions for a large variety of x values tends to compromise immense portions of the FPGA storage space, and thus it is prohibitive. To overcome this issue, the mentioned functions were split into line segments given by:

$$\tilde{f}(x) = mx + n, \text{ for } x_a \leq x < x_b \quad (1)$$

where $[x_a, x_b]$ is the interval for which $\tilde{f}(x)$ is valid. Thus, instead of storing the different values of $f(x)$, we can table the values of m and n for the different segments of the function. This approach requires some extra computations, say it $\tilde{f}(x) = mx + n$, but significantly reduces the required storage space.

In this case, coefficients for the line segment in the interval $[x_a, x_b]$ is given by:

$$m = \frac{f(x_b) - f(x_a)}{x_b - x_a} \quad (2)$$

$$n = f(x_a) - mx_a \quad (3)$$

2 Computing the error for the LUT for $f(x) = \sqrt{x}$

$$\sqrt{x} \approx \tilde{f}_p(x) = m_p x + n_p \quad (4)$$

where

$$m_p = \frac{\sqrt{x_{p+1}} - \sqrt{x_p}}{x_{p+1} - x_p} \quad (5)$$

$$n_p = \sqrt{x_p} - m_p x_p \quad (6)$$

$$\mathcal{Err} = \frac{\sqrt{x} - \tilde{f}_p(x)}{\sqrt{x}} \quad (7)$$

$$= \frac{\sqrt{x} - (m_p x + n_p)}{\sqrt{x}} \quad (8)$$

$$= \frac{\sqrt{x} - m_p x - (\sqrt{x_p} - m_p x_p)}{\sqrt{x}} \quad (9)$$

$$= \frac{\sqrt{x} - \sqrt{x_p} + m_p(x_p - x)}{\sqrt{x}} \quad (10)$$

Let $\Delta x_p = x_{p+1} - x_p$ and $\Delta x = x - x_p$, then $x = x_p + \Delta x$:

$$m_p = \frac{\sqrt{x_p + \Delta x_p} - \sqrt{x_p}}{\Delta x_p} \quad (11)$$

$$\mathcal{E}_{rr} = \frac{\sqrt{x_p + \Delta x} - \sqrt{x_p}}{\sqrt{x_p + \Delta x}} - \frac{m_p \Delta x}{\sqrt{x_p + \Delta x}} \quad (12)$$

$$\mathcal{E}_{rr} = 1 - \frac{\sqrt{x_p}}{\sqrt{x_p + \Delta x}} - \frac{m_p \Delta x}{\sqrt{x_p + \Delta x}} \quad (13)$$

where

$$m_p = \frac{\sqrt{x_p + \Delta x_p} - \sqrt{x_p}}{\Delta x_p} \quad (14)$$

3 Computing the error for the LUT for $f(x) = \mathbf{tg}^{-1}(x)$

$$\arctan(x) \approx \tilde{f}_p(x) = m_p x + n_p \quad (15)$$

where

$$m_p = \frac{\arctan(x_{p+1}) - \arctan(x_p)}{x_{p+1} - x_p} \quad (16)$$

$$n_p = \arctan(x_p) - m_p x_p \quad (17)$$

$$\mathcal{E}_{rr} = \frac{\arctan(x) - \tilde{f}_p(x)}{\arctan(x)} \quad (18)$$

$$= \frac{\arctan(x) - (m_p x + n_p)}{\arctan(x)} \quad (19)$$

$$= \frac{\arctan(x) - m_p x - (\arctan(x_p) - m_p x_p)}{\arctan(x)} \quad (20)$$

$$= \frac{\arctan(x) - \arctan(x_p) + m_p(x_p - x)}{\arctan(x)} \quad (21)$$

Let $\Delta x_p = x_{p+1} - x_p$ and $\Delta x = x - x_p$, then $x = x_p + \Delta x$:

$$m_p = \frac{\arctan(x_p + \Delta x_p) - \arctan(x_p)}{\Delta x_p} \quad (22)$$

$$\mathcal{E}_{rr} = \frac{\arctan(x_p + \Delta x) - \arctan(x_p)}{\arctan(x_p + \Delta x)} - \frac{m_p \Delta x}{\arctan(x_p + \Delta x)} \quad (23)$$

$$\mathcal{E}_{rr} = 1 - \frac{\arctan(x_p)}{\arctan(x_p + \Delta x)} - \frac{m_p \Delta x}{\arctan(x_p + \Delta x)} \quad (24)$$

where

$$m_p = \frac{\arctan(x_p + \Delta x) - \arctan(x_p)}{\Delta x} \quad (25)$$